

1st sit COURSEWORK Question Paper:

Spring Semester 2026

Module Code:	CS4051NI/CC4059NI
Module Title:	Fundamentals of Computing
Module Leader:	Pratik Panta (Islington College)

Coursework Type:	Individual
Coursework Weight:	This coursework accounts for 70% of your total module grades.
Submission Date:	Milestone: Week 10 Friday, April 24, 2026 Final Submission: Week 12 Friday, May 08, 2026
When Coursework is given out:	Week 6
Submission Instructions:	Submit the following to Islington College MST Portal before 06:00 PM on the due date: • Soft copy of the report (.pdf format) • Zip file with source code of the program in google drive • Drive link in cover page of the report
Warning:	London Metropolitan University and Islington College Kathmandu take Plagiarism seriously. Offenders will be dealt with sternly.

PLAGIARISM

You are reminded that there are regulations concerning plagiarism. Extracts from these regulations are printed overleaf. Please sign below to say that you have read and understand these extracts:

Extracts from University *Regulations on Cheating, Plagiarism and Collusion*

Section 2.3: *“The following broad types of offence can be identified and are provided as indicative examples*

- . *Cheating: including taking unauthorized material into an examination; consulting unauthorized material outside the examination hall during the examination; obtaining an unseen examination paper in advance of the examination; copying from another examinee; using an unauthorized calculator during the examination or storing unauthorized material in the memory of a programmable calculator which is taken into the examination; copying coursework.*
- . *Falsifying data in experimental results.*
- . *Personation, where a substitute takes an examination or test on behalf of the candidate. Both candidate and substitute may be guilty of an offence under these Regulations.*
- . *Bribery or attempted bribery of a person thought to have some influence on the candidate’s assessment.*
- . *Collusion to present joint work as the work solely of one individual.*
- . *Plagiarism, where the work or ideas of another are presented as the candidate’s own.*
- . *Other conduct calculated to secure an advantage on assessment.*
- . *Assisting in any of the above.*

Some notes on what this means for students:

1. Copying up another student's work is an offence, whether from a copy on paper or from a computer file, and in whatever form the intellectual property being copied takes, including text, mathematical notation, and computer programs.
2. Taking extracts from published sources *without attribution* is an offence. To quote ideas, sometimes using extracts, is generally to be encouraged. Quoting ideas is

achieved by stating an author's argument and attributing it, perhaps by quoting, immediately in the text, his or her name and year of publication, e.g. " $e = mc^2$ (Einstein 1905)". A *reference* section at the end of your work should then list all such references in alphabetical order of authors' surnames. (There are variations on this referencing system which your tutors may prefer you to use.) If you wish to quote a paragraph or so from published work then indent the quotation on both left and right margins, using an italic font where practicable, and introduce the quotation with an attribution.

School of Computing, FLSC

Contract cheating

Contract cheating (also known as assessment outsourcing, commissioning or ghost writing) is when someone seeks out another party, or AI generator service, to produce work or buy an essay or assignment, either already written or specifically written for them or the assignment to submit as their own piece of work.

Contract cheating undermines the integrity of the academic process and devalues the qualifications awarded by the university. Students are reminded that academic integrity is a fundamental principle of our institution. Engaging in contract cheating not only impacts the individual's academic record but also the reputation of the university.

Students are encouraged to seek support if they are struggling with their coursework. The university offers a range of resources, including academic counseling, tutoring services, and workshops on study skills and time management. Utilizing these resources can help students achieve their academic goals without resorting to dishonest practices.

Penalty:

- Failure in the Module: The student must re-register for the same module, and the re-registered module will be capped at a bare pass.
- Ineligibility to Continue on the Course: Where re-registration of the same module, or a suitable alternative, is not permissible, the student will not be able to continue on the course. Additionally, the following penalty will be applied to the student's final award:
 - Undergraduate Honors: The student's final classification will be reduced by one level.
 - Unclassified Bachelors: Downgraded to Diploma in Higher Education.

- Foundation Degree: Distinction downgraded to Merit; Merit downgraded to Pass; Pass downgraded to Certificate in Higher Education.
- Masters: Distinction downgraded to Merit; Merit downgraded to Pass; Pass downgraded to Postgraduate Diploma.

Reporting and Consequences:

Instances of contract cheating will be thoroughly investigated, and students found guilty will face the penalties outlined above. It is the responsibility of every student to ensure that their work is their own and to avoid situations that could lead to accusations of academic misconduct.

By adhering to these standards, students contribute to a fair and equitable academic environment, ensuring the value and recognition of their qualifications are maintained.

This module is assessed by coursework (70%). For the coursework, the students are required to develop an application based on detailed guidance on given specifications. Through the coursework students should be able to:

- ✓ Develop a ***MedStore Wholesale System***
 - ✓ Describe the program
 - ✓ Test the program with some sample data to demonstrate its behavior
 - ✓ Write a report to present the work
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- Students are required to submit weekly assignments along with related research evidence
 - The guideline for the Documentation/Development part is given below:
 - Components to be included are cover page, and table of contents, figures and tables, footer.
 - The report must have an **Introduction** section with definite goals and objectives (approx. 400 words or more).

The report must have a **Discussion and Analysis** section in which students need to explain how the program was developed.

1. Scenario

MedStore Pvt. Ltd. is a wholesale distributor that manages stock for a wide range of medicines across different brands. The store maintains bulk quantities of medicines and allows sales either per **tablet** or per **strip** (a strip containing a fixed number of tablets). As a wholesaler, the company encourages customers to purchase in bulk and provides a discount on strip sales, offering **5% off when two or more strips of the same medicine are purchased**. The company wants to keep track of both sales transactions and stock updates in real time. For this purpose, it maintains the details of the medicines in a text file. With each transaction, whether a sale to a customer or a restock from a supplier, an invoice or note must be generated to record the details.

An application needs to be developed which will read the **medicine inventory text file** and display all the medicines available in stock. With each transaction, a note or invoice should be generated for the customer and written in a file, including details such as the medicine name, brand, quantity purchased, unit type (tablet or strip), applicable discount, and the total amount. The stock of the medicines must also be updated in the inventory file immediately after each transaction so that the system always reflects the latest availability.

A sample format of the text file including the information about the medicine wholesale system is as follows:

Paracetamol 500mg, Lomus, 1200, 5, 45, 10

Cetirizine 10mg, Quest, 800, 4, 35, 12

Amoxicillin 500mg, Nepal Remedies, 500, 12, 110, 15

Pantoprazole 40mg, Deurali-Janta, 600, 7, 60, 8

ORS Sachet, Time Pharma, 300, 20, 180, 12

1st column shows the medicine name, 2nd column shows the brand name, 3rd column shows the quantity available in stock (in tablets), 4th column shows the rate per tablet in rupees, 5th column shows the rate per strip in rupees, and 6th column shows the number of tablets in one strip.

Note: You can use your own format and add other information too.

A note/invoice should be generated for each transaction. When a medicine is sold, a note/vat invoice should be generated (as a .txt file) which must contain the medicine name, brand, unit type (tablet or strip), quantity sold, name of the customer, date of transaction, the discount amount (if applicable), and the total cost. If a customer decides to buy more than one type of medicine, then the total should be calculated by adding up all the purchased items, applying the strip discount properly wherever applicable.

When the company wants to restock the medicines from the suppliers, a note/vat invoice should be generated which should include the medicine name, brand, quantity bought (in tablets or strips), rate of the medicine, name of the vendor/supplier, date of transaction, and the total amount. If restock includes multiple types of medicines, a single invoice should be created for each transaction. The items should also be updated in the original inventory file on every stock.

The format of the notes/invoices is up to you. But each file should have a unique name.

2. Algorithm

- An algorithm should be developed for the application where everything the program does should be considered. The algorithm should be described in steps, pseudocode, and flowcharts should also be included.

3. Data Structures

- The programming should be done using data structures and operations in Python for input/output, character and string processing, and data storage.
- It can use any primitive or complex data structures which might be necessary for holding the data (pairs, lists, strings, dictionaries, etc.)
- The choice of data structures must be specified in the report.

4. Program

- The program must work in a loop, displaying the available lands and waiting for the administrator to enter details of the customers. The program should not close unless the administrator decides to do so.
- The program must check the input data, displaying error messages whenever unwanted data is entered, for example if some string value is entered where a numerical value is expected.
- The program must be implemented in a modular way with separate functions for the different operations such as input/output, reading files, generating invoices/notes, etc.

5. Description

- The program must be described in terms of its structure and behavior.
- It can be presented using text and structural charts, flowcharts or other diagrams as needed.
- The report must have a **Testing** section with evidence (through appropriate screenshots) that has been carried out for the program.
- The report must include a **Conclusion** section where they need to present their findings of the development and research (approx. 300 words).
- Demonstration of the project.
- If any individual student cannot justify his/her project, then the project will be kept under plagiarism.

NOTE: The technicality of the project will be judged during the demo and marked accordingly.

Marking Breakdown for Coursework	
Criteria	Total Marks
1. Introduction	5
2. Algorithm	5
3. Pseudocode	5
4. Flowchart	5
5. Data Structures	5
6. Program	
a. Correctness	10
b. Implementation (modularity, use of functions/classes)	10
c. Programming Style	10
d. Exception Handling	10
e. User interface/program usability	10
7. Testing	5
8. Conclusion	5
9. VIVA	10
10. Report Structure and Formatting	5
Total:	100